

1 Literature Study Table

Title of Paper	Methods Used	Features	Advantages	Disadvantages	Future Scope
A Novel Approach for Crowd Analysis and Density Estimation by using ML Techniques	- MobileNet SSD, - YOLO v4 - Mask-RCNN	- Lightweight and versatile - Adaptable to real-time processing	- MobileNet SSD: Lightweight, versatile, devices with limited computational power. - YOLO v4: High precision, real-time processing - Mask-RCNN: Excellent instance segmentation, contextual	- MobileNet SSD: Reduced accuracy in high-density crowds. - YOLO v4: High computational demand. - Mask-RCNN: Slow processing, computationally heavy	Real-time optimization for dense crowds
A Proposal on Stampede Detection in Real Environments	- Optical flow computation - feature extraction, in real time. - Stacking classifiers to detect stampedes	- No thresholds needed - Public dataset use	- Real-time detection - Public dataset robustness - Doesn't require predefined thresholds. - Stacking classifier enhances performance. - Public datasets ensure robustness.	- Limited performance in high-density crowds - Limited performance in high-density crowds. - Relies on optical flow techniques. - Applicable only to a small number of scenarios.	Further research for high-density crowds

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CrowdFormer: An Overlap Patching Vision Transformer	- Weakly-supervised crowd counting model with tranformer-encoder and regression - Vision Transformer (ViT) model for crowd analysis. - Maintains spatial continuity & reduces information loss.	- Preserves long-term dependencies using overlapping patches - Improves performance on crowd density estimation and localization tasks. - Employs Self Attention Mechanism	- Reduces information loss from non-overlapping patches. - Preserves local and global contextual information. - Enhances accuracy while maintaining efficiency.	- Increased computational complexity compared to traditional ViTs. - Requires careful hyperparameter tuning for optimal performance.	Application in real-time applications and hybrid models.
PVT v2: Improved Baselines with Pyramid Vision Transformer	- Improvement of PVT v1 - Employs Self Attention Mechanism - Vision Transformer (ViT) model for crowd analysis. - Maintains spatial continuity & reduces information loss.	- Linear spatial reduction attention (LSRA) layer - Overlapping patch embedding (OPE) scheme, - Convolutional feed-forward network (CFFN)	- LSRA significantly lowers computational complexity & makes model more efficient. - Overlapped patch embedding enhances the extraction of local features, improves the overall representation. - CFFN contributes to robust feature learning without fixed positional embeddings.	- Increased computational complexity compared to traditional ViTs. - Requires careful hyperparameter tuning for optimal performance.	Integration with additional local operations or adaptations to other domains
Machine Learning for Dense Crowd Direction Prediction Using LSTM	- LSTM model predicts movement trajectories based on past positions. - Cone of vision concept for influence detection. "Unit Angle" quantifies these neighbors.	- Trajectory prediction - Cone of vision concept	- Suitable for both structured and unstructured crowds. - Retains long-term dependencies for more accurate predictions. - Improves trajectory prediction for individuals in crowds.	- Requires careful tuning for optimal performance. - Expensive requiring high-quality sequential data.	Real-world testing required due to complexity.

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CvT: Introducing Convolutions to Vision Transformers	◦ Convolutional Vision Transformers (CvT), which incorporate convolutional layers into the ViT framework ◦ Replaces standard self-attention layers with convolutional token embedding and projection mechanisms	◦ Trajectory prediction ◦ Hierarchical feature representation to reduce computational overhead	◦ Improves feature locality using convolutional embedding. ◦ Reduces computational complexity compared to standard ViTs. ◦ Achieves superior performance on image classification tasks.	◦ Still computationally expensive compared to traditional CNNs. ◦ Requires careful tuning of convolutional and attention components.	◦ Further optimization is needed for real-time applications
Pyramid Vision Transformer: A Versatile Backbone for Dense Prediction	◦ Novel Transformer backbone that incorporates a hierarchical structure ◦ Pyramid structure that progressively reduces the spatial dimensions of feature maps while increasing channel depth	◦ Spatial-reduction attention mechanism to mitigate high memory consumption ◦ Multi-stage processing to refine feature representations at different scales ◦ PVT outperforms CNN-based backbones while maintaining efficiency	◦ Incorporates a hierarchical feature representation similar to CNNs. ◦ Reduces memory and computational overhead via spatial-reduction attention. ◦ Improves performance on dense prediction tasks such as segmentation and object detection.	◦ Requires substantial computational resources compared to traditional CNNs. ◦ Transformer-based models generally need large-scale training datasets for optimal performance.	◦ Further optimizations are needed to enhance real-time processing capabilities ◦ Improve efficiency in resource-constrained environments
Jhu-Crowd++: Large-Scale Crowd Counting Dataset and Benchmark Method	◦ Large-scale dataset ◦ Benchmark methods	◦ Diverse scenarios ◦ Rich annotations	◦ Diverse scenarios and environmental conditions. ◦ Rich annotations at both image and head level. ◦ Distractor images.	◦ Annotation complexity ◦ Limited diversity in distractors. ◦ Limited focus on temporal data.	◦ Exploration of temporal data needed ◦ Generalization to unseen scenarios ◦ Enhancing the dataset diversity and leveraging multi-camera systems.

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Background Subtraction in Real Applications: Challenges, Current Models and Future Directions	◦ Gaussian Mixture Models (GMM): Probabilistic approach to model the background and adapt to changes over time. ◦ ViBe Algorithm	◦ Efficiently separates objects in motion from a static background. ◦ Pixel-based technique that maintains a history of background values and updates dynamically. ◦ Combination with CNN, RNN improve background-foreground segmentation.	◦ GMM provides adaptability to illumination changes and dynamic backgrounds. ◦ Computationally efficient and robust against sudden background variations. ◦ Accuracy in complex environments with occlusions and dynamic elements.	◦ GMM can be sensitive to sudden lighting changes and requires parameter tuning. ◦ ViBe may struggle with highly dynamic scenes and complex movements.	◦ Optimization for real-time applications ◦ Optimization for minimal computational overhead
Transformers in Time Series: A Survey	◦ Sparse Attention Mechanisms ◦ Alternative positional encoding schemes tailored for continuous data. ◦ Creation of Hybrid Models modeling. ◦ Uses windowing and memory-efficient attention mechanisms to scale to long time series.	◦ Hybrid modelling with convolutional networks (CNN) improves strength in sequential data ◦ Capable of capturing long-range dependencies in time series data. ◦ Handles multi-modal & irregular time series data more effectively than traditional models.	◦ Reduced quadratic complexity of standard self-attention. ◦ Advances in sparse attention make Transformers more computationally efficient for time series applications.	◦ High computational complexity, especially for long sequences. ◦ Requires large-scale training data to generalize well. ◦ Model interpretability remains a challenge compared to traditional statistical methods. ◦ Hyperparameter tuning can be complex.	◦ Optimization for real-time applications ◦ Optimization for minimal computational overhead ◦ Improve interpretability
A CNN-RNN Neural Network Join LSTM for Crowd Counting	◦ CNN for feature extraction & ◦ LSTM for context relevance to estimate crowd density	◦ Estimates crowd density	◦ High accuracy in dense settings ◦ Robust across densities	◦ Poor performance in sparse crowds	◦ Optimization needed for sparse environments